# Uniform families in the shape of the D = 8���10 Haar records

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## Uniform families in the shape of the D = 8–10 Haar records: the LPS family, a 4/3 obstruction, and what a cyclic factor does

Doğa Özden's degree–diameter lab, with Claude Code (job `thm-haar-families`, 2026-10-27). Written for Gabriela Araujo-Pardo and Ignacio López. Draft; corrections welcome.

**Labels.** Every statement carries one of three labels.

- **Theorem, Proposition, Lemma, Corollary**: complete proof in this note, or a cited result named in place (marked cited, with the source and how we know it: **[read]** primary source read, **[abstract]** abstract only, **[secondary]** known from a book, survey or citation).
- **Computer-checked**: a statement about finitely many explicit graphs, verified by programs that are named, with logs.
- **Heuristic**: a model, a measurement or an interpretation. Not used in any proof.

**The question.** The lab's records at D = 8, 9, 10 (`research/theorems/HAAR_D8_10.md` on branch `claude/thm-haar-d8-10-sf7f0a`, cited below as [H]) are Haar graphs of SL(2, q) × $Z_m$ or PSL(2, q) × $Z_m$ whose connection set S is closed under an automorphism of order 3, each found by computer search for one group. Can they be turned into infinite families, with S given by one formula in q (and m) and a proved diameter?

**The answer, stated plainly.**

1. There is an explicit infinite family of exactly this shape, with m = 1: the connection sets of Lubotzky, Phillips and Sarnak (LPS). For every odd prime p and every prime q > p, the p + 1 LPS matrices in PGL(2, q) are permuted by conjugation with an element of order 3 of PSL(2, q), with cycle type 1+1+3+⋯+3 or 3+⋯+3: the cycle types that the orbit search used at Δ = 6, 8, 12, 14 (Theorem 2.6). Their diameter is at most 2 log_p n + O(1) (cited, LPS).
2. This family provably cannot reach the records. The vertex ψ(i) is at distance exactly the least even L with $2p^{L/2} > q^2$ (Theorem 3.2: lower bound for every odd p, equality for p ≡ 1 mod 4), so the diameter is at least (4/3) log_p n − 2 log_p 2 (Corollary 3.3, the bound of N. T. Sardari, here with a two-paragraph proof), and a member of diameter ≤ D has at most $\sqrt{2}\cdot p^{1-D/4}$ of the bipartite Moore bound (Corollary 3.4). Computed: the LPS graphs are 2.4 to 15 times below the 2013 table at D = 8–10 (Appendix B).
3. A cyclic factor removes this obstruction in practice, but not as a formula. The obstruction survives every cover (Proposition 4.1), but it bounds q, not m. LPS × $Z_m$ with labels chosen for each (q, m) by a squad reaches 99.4 % of the table at (6,10) and 98.9 % at (8,8) (Computer-checked 4.2), still below the lab's records.
4. Other uniform formulas behave like one random draw. Integer lifts, primitive-root and Dickson formulas, the conic model and torus orbits (18 squad cards, §5) give an excess diam − $log_{\Delta-1}|G|$ of about 3.0–3.4 at Δ = 6–8, the same as a single random orbit-closed S, against about 2.0–2.3 for the records. Heuristically the records' advantage is selection among millions of sets, which no single formula provides (§5–6).
5. What would be needed is open. A uniform family of record quality at fixed Δ would be an explicit family of bounded-degree graphs with diameter (1 + o(1)) $log_{\Delta-1}$ n. None is known; for the LPS graphs Sardari proved that it fails (§6).

No infinite family above the table is proved in this note.

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Appendix A. The squad rounds. Appendix B. The LPS family cell by cell.


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## 0. Summary of statements

| # | statement | label |
|---|---|---|
| Prop. 1.1 | at fixed (Δ, D) only finitely many groups; for Q × $Z_m$ with Q perfect, m ≤ A(Δ, D) | Proposition (from [H]) |
| Lemma 2.3 | ψ: H(Z) → $M_2(F_q)$ is a ring homomorphism with kernel qH(Z); det ψ(α) = N(α); ψ(ᾱ) = adj ψ(α) | Lemma |
| Lemma 2.4 | for q > p the p + 1 matrices ψ(α), α ∈ $S_p$, are distinct in PGL(2, q), and ψ($S_p$) is closed under inversion | Lemma |
| Lemma 2.5 | structure of the component $\Gamma_{p,q}$; diam $\Gamma_{p,q}$ = ecc(I, A) | Lemma |
| Thm. 2.6 | conjugation by h = ψ(1+i+j+k), of order 3 in PSL(2, q), permutes ψ($S_p$) with cycle type 1+1+$3^{(p-1)/3}$ (p ≡ 1 mod 3), $3^{(p+1)/3}$ (p ≡ 2 mod 3), 1+3 (p = 3) | Theorem |
| Thm. 2.7 | $\Gamma_{p,q}$ is connected and Ramanujan; diam ≤ 2 log_p n + O(1) | cited (LPS) |
| Lemma 3.1 | an even walk from (I, A) ends at ψ(β) with N(β) = $p^L$ and β ≡ 1 mod 2H(Z) | Lemma |
| Thm. 3.2 | d((I,A), (ψ(i),A)) ≥ $L_0$ := min{L even : $2p^{L/2} > q^2$}, with equality if p ≡ 1 mod 4 | Theorem (equality uses a cited factorization) |
| Cor. 3.3 | diam $\Gamma_{p,q}$ > (4/3) log_p n − 2 log_p 2 | Corollary |
| Cor. 3.4 | if diam $\Gamma_{p,q}$ ≤ D then n < $2^{3/2}$ $p^{3D/4}$ and n < √2·$p^{1-D/4}$·$M^b$(p+1, D) | Corollary |
| 3.5 | equality in Thm. 3.2 for all 8 primes p ≤ 29 tested (including p ≡ 3 mod 4) and all primes p < q ≤ 160; ecc − $L_0$ ∈ {0, 1, 2} | Computer-checked |

| # | statement | label |
|---|---|---|
| Prop. 4.1 | the bound of Thm. 3.2 holds in every cover H(PGL(2,q) × K, S̃) of the LPS graph | Proposition |
| 4.2 | LPS × $Z_m$ with searched labels: 99.4 % of the table at (6,10), 98.9 % at (8,8) | Computer-checked |
| §5 | uniform formulas of six kinds: excess about that of one random draw | Computer-checked (scores), Heuristic (reading) |

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## 1. Setting, notation and projective coordinates

### 1.1 Haar graphs

We use the conventions of [H] §1–2. For a finite group G and S ⊆ G with |S| = Δ, the Haar graph H(G, S) has vertex set G × {A, B}, and (g, A) is adjacent to (sg, B) for every s ∈ S. It is bipartite and Δ-regular, the right translations $\rho_h$(g, X) = (gh, X) are automorphisms acting regularly on each side ([H] Lemma 1.2), and its diameter is max(ecc(e, A), ecc(e, B)) ([H] Cor. 2.3). A walk of length L from (e, A) uses letters $t_1$, ..., $t_L$ ∈ S and ends at (w, A) or (w, B) with

$$w = \cdots t_4^{-1}\, t_3\, t_2^{-1}\, t_1 \qquad (\text{an } A\text{-word}),$$

and the walk backtracks exactly when two consecutive letters are equal ([H] Prop. 2.2). Lemma 3.1 of [H]: H(G, S) is isomorphic to H(G, aψ(S)$^{\pm 1}$b) for a, b ∈ G and ψ ∈ Aut(G); the map (g, A) ↦ (g, B), (h, B) ↦ (h, A) is an isomorphism H(G, S) → H(G, S⁻¹).

Throughout, $M^b(\Delta, D) = 2\,\Sigma_{j=0}^{D-1}\,(\Delta-1)^j$ is the bipartite Moore bound, and n denotes the number of vertices of the graph at hand.

### 1.2 Projective coordinates

Let q be an odd prime. An element of PGL(2, q) is a nonsingular 2 × 2 matrix up to a nonzero scalar, that is, a point

$$[X] = [a : b : c : d] \in \mathrm{PG}(3, q), \qquad X = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

not on the quadric $\mathcal{Q}$ : ad − bc = 0. The quadric $\mathcal{Q}$ is a hyperbolic quadric Q⁺(3, q) with (q + 1)² points, so PGL(2, q) is the set of the (q + 1)(q² + 1) − (q + 1)² = q(q² − 1) points of PG(3, q) off $\mathcal{Q}$. Whether ad − bc is a square does not depend on the representative, and PSL(2, q) is the set of the q(q² − 1)/2 points with ad − bc a nonzero square. For M ∈ GL(2, q), left multiplication [X] ↦ [MX] is a collineation of PG(3, q) that fixes $\mathcal{Q}$ (det MX = det M det X).

So a Haar graph H(PGL(2, q), S) is the bipartite graph on two copies of PG(3, q) $\setminus\ \mathcal{Q}$ in which **the point [X] of the first copy is joined to the points [MX], M ∈ S, of the second copy**; and the point [Y] of the second copy is joined to the points [adj(M)·Y], M ∈ S, of the first, where adj [[a, b], [c, d]] = [[d, −b], [−c, a]] is the inverse up to a scalar. These are the exact neighbourhoods of every vertex, once S is written down (§2, Example 2.8).

### 1.3 The records' shape

[H] Construction 3.2: S is closed under an automorphism α of order k, here k = 3, and splits into f fixed letters and 3-cycles; for G = Q × $Z_m$ the automorphism is α(x, z) = (gxg⁻¹, rz) with g of order 3 modulo the centre of Q and r of multiplicative order 3 mod m. The cycle types of the records are in [H] §5.1: 3+3 at (6,9), (6,10);

1+3+3 at Δ = 7; 1+1+3+3 at (8,8), (8,9), (8,10); 3+3+3 at (9,8), (9,9), (9,10); 3+3+3+3 at (12,9), (12,10); 1+1+3+3+3+3 at (14,9), (14,10); and so on. Since |S| = f + 3c, f ≡ Δ (mod 3); every record with k = 3 uses the least such f ∈ {0, 1, 2}.

By a **uniform family** we mean, informally, one rule that produces S from q (and m) without a search for each q: a formula, as for the Singer difference sets at D = 3. All the families below are uniform in this sense, except where a search over labels is stated.

### 1.4 Where a uniform family can live

**Proposition 1.1.** (a) For fixed Δ ≥ 2 and D, only finitely many groups G admit S with |S| = Δ and diam H(G, S) ≤ D. (b) If G = Q × $Z_m$ with Q perfect (for example Q = SL(2, q) or PSL(2, q), q ≥ 4) and diam H(G, S) ≤ D, then m ≤ A(Δ, D) = Θ($D^{\Delta-1}$) for fixed Δ.

Proof. (a) |G| ≤ $M^b$(Δ, D)/2 ([H] Thm. 4.1). (b) [H] Cor. 4.4(ii). ■

So an infinite uniform family of the records' shape has Δ fixed and D → ∞ (and then q → ∞, since m grows at most polynomially in D while |G| must grow exponentially), or D fixed and Δ → ∞. The LPS family below can be read in both regimes (Corollary 3.4).

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## 2. The LPS family has the records' shape

### 2.1 Definitions

Let H(Z) = {$a_0 + a_1 i + a_2 j + a_3 k$ : $a_t \in Z$} be the Lipschitz quaternions ($i^2 = j^2 = k^2 = ijk = -1$), with conjugate $\bar{\alpha} = a_0 - a_1 i - a_2 j - a_3 k$ and norm $N(\alpha) = \alpha\bar{\alpha} = a_0^2 + a_1^2 + a_2^2 + a_3^2$. N is multiplicative.

**Definition 2.1 (the LPS set).** For an odd prime p let $S_p$ be the set of α ∈ H(Z) with N(α) = p and

- if p ≡ 1 (mod 4): $a_0 > 0$ odd, and $a_1$, $a_2$, $a_3$ even;
- if p ≡ 3 (mod 4): $a_0$ even, $a_1$, $a_2$, $a_3$ odd, and either $a_0 > 0$, or $a_0 = 0$ and $a_1 > 0$.

By Jacobi's four-square theorem $|S_p| = p + 1$ (cited: LPS, Combinatorica 8 (1988), §2, and G. Davidoff, P. Sarnak, A. Valette, Elementary Number Theory, Group Theory and Ramanujan Graphs, LMS Student Texts 55 (2003), Ch. 2 [secondary]). If α ∈ $S_p$ then $\bar{\alpha}$ or $-\bar{\alpha}$ is in $S_p$.

**Definition 2.2 (the splitting).** Let q be an odd prime, q ≠ p. Choose x, y ∈ $F_q$ with $x^2 + y^2 + 1 = 0$ (they exist: the sets $\{x^2\}$ and $\{-1 - y^2\}$ both have (q + 1)/2 elements, so they meet). Put

$$\psi(a_0 + a_1 i + a_2 j + a_3 k) = \begin{pmatrix} a_0 + a_1 x + a_3 y & -a_1 y + a_2 + a_3 x \\ -a_1 y - a_2 + a_3 x & a_0 - a_1 x - a_3 y \end{pmatrix} \bmod q.$$

**Lemma 2.3.** ψ: H(Z) → $M_2(F_q)$ is a ring homomorphism with kernel qH(Z), and it is onto. det ψ(α) ≡ N(α), and ψ($\bar{\alpha}$) = adj ψ(α), so ψ(α)ψ($\bar{\alpha}$) = N(α)·I.

Proof. ψ is additive and ψ(1) = I. Put $M_i$ = ψ(i) = [[x, −y], [−y, −x]], $M_j$ = ψ(j) = [[0, 1], [−1, 0]], $M_k$ = ψ(k) = [[y, x], [x, −y]]. Direct multiplication gives $M_i{}^2 = (x^2 + y^2)\cdot I = -I$, $M_j{}^2 = -I$, $M_i \cdot M_j = M_k$ and $M_j \cdot M_i$ = [[−y, −x], [−x, y]] = $-M_k$. Since H(Z) is the ring generated over Z by i, j subject to $i^2 = j^2 = -1$ and ji = −ij (with k = ij), the assignment i ↦ $M_i$, j ↦ $M_j$ extends to a ring homomorphism, and it sends k to $M_i \cdot M_j = M_k$; this is the displayed formula. Its kernel contains qH(Z). If ψ(α) = 0, the diagonal gives $a_0 + a_1 x + a_3 y = a_0 - a_1 x - a_3 y = 0$, so $a_0 = 0$ and $a_1 x + a_3 y = 0$; the off-diagonal gives $a_2 = 0$ and $a_3 x - a_1 y = 0$. Then $a_1(x^2 + y^2) = x(a_1 x + a_3 y) - y(a_3 x - a_1 y) = 0$, so $a_1 = 0$ ($x^2 + y^2 = -1$), and then $a_3 x = a_3 y = 0$ with (x, y) ≠ (0, 0), so $a_3 = 0$, all mod q. So the kernel is qH(Z), and the induced map H(Z)/qH(Z) → $M_2(F_q)$ is injective between sets of size $q^4$, hence onto. The determinant of the displayed matrix is $(a_0 + a_1 x + a_3 y)(a_0 - a_1 x - a_3 y) - (a_2 + a_3 x - a_1 y)(-a_2 + a_3 x - a_1 y) = a_0^2 - (a_1 x + a_3 y)^2 + a_2^2 - (a_3 x - a_1 y)^2 = a_0^2 + a_2^2 - (a_1^2 + a_3^2)(x^2 + y^2) = N(\alpha)$. Finally adj ψ(α) has diagonal $(a_0 - a_1 x - a_3 y, a_0 + a_1 x + a_3 y)$ and off-diagonal entries $-(-a_1 y + a_2 + a_3 x)$, $-(-a_1 y - a_2 + a_3 x)$, which is ψ($a_0 - a_1 i - a_2 j - a_3 k$). ■

**Lemma 2.4.** Let q > p. (a) If α, β ∈ H(Z) have N(α) = N(β) = p and ψ(α) = cψ(β) for some c ∈ $F_q$^×, then α = ±β. (b) The p + 1 points [ψ(α)], α ∈ $S_p$, of PGL(2, q) are distinct, and the set they form is closed under inversion.

Proof. (a) γ = αβ has N(γ) = p², and ψ(γ) = ψ(α)ψ(β) = cψ(β)ψ(β) = cp·I. By Lemma 2.3, γ − c̃p ∈ qH(Z) for an integer c̃ ≡ c, so the coordinates γ₁, γ₂, γ₃ are divisible by q. They are at most √N(γ) = p < q in absolute value, so they vanish: γ is an integer with γ² = p², γ = ±p. Then α = γβ⁻¹ = ±pβ/N(β) = ±β. (b) Distinct elements of $S_p$ are not negatives of each other (the sign conditions of Definition 2.1 fix the sign), so (a) gives distinctness. By Lemma 2.3, [ψ(α)]⁻¹ = [adj ψ(α)] = [ψ(ᾱ)] = [ψ(±ᾱ)], and ±ᾱ ∈ $S_p$. ■

**Definition 2.5′.** $\Gamma_{p,q}$ is the connected component of (I, A) in H(PGL(2, q), ψ($S_p$)).

**Lemma 2.5 (structure of $\Gamma_{p,q}$).** Let q > p and let c ∈ ψ($S_p$).

(a) All of ψ($S_p$) lies in one coset of PSL(2, q): in PSL(2, q) if p is a square mod q, in the other coset if not.
(b) The A-vertices of $\Gamma_{p,q}$ form $H_0$ × {A} for a subgroup $H_0$ ≤ PSL(2, q), and its B-vertices c$H_0$ × {B}. The right translations $\rho_h$, h ∈ $H_0$, map $\Gamma_{p,q}$ to itself and act regularly on each side. When $H_0$ = PSL(2, q), $\Gamma_{p,q}$ has n = q(q² − 1) vertices.
(c) diam $\Gamma_{p,q}$ = ecc(I, A) (computed in $\Gamma_{p,q}$).
(d) $\Gamma_{p,q}$ ≅ H(PSL(2, q), c⁻¹ψ($S_p$)) when it has q(q² − 1) vertices.

Proof. (a) det ψ(α) = p for all α (Lemma 2.3), and PSL(2, q) is the kernel of the homomorphism PGL(2, q) → $F_q$^×/($F_q$^×)² induced by the determinant. (b) A letter and its inverse both have determinant class [p], so an A-word of length L has class $[p]^L$: it lies in PSL(2, q) for L even and in cPSL(2, q) for L odd. The set $H_0$ of the values of even A-words (the A-side of $\Gamma_{p,q}$) is closed under products and contains I, so it is a subgroup of the finite group PSL(2, q). Right translation $\rho_h$ by h ∈ $H_0$ is an automorphism ([H] Lemma 1.2) mapping (I, A) to (h, A) ∈ $\Gamma_{p,q}$, so it maps $\Gamma_{p,q}$ to itself; {$\rho_h$ : h ∈ $H_0$} acts regularly on the A-side $H_0$ × {A} and on the B-side c$H_0$ × {B}. If $\Gamma_{p,q}$ is connected in the sense of Theorem 2.7, $H_0$ = PSL(2, q). (c) By (b) and [H] Lemma 2.1 (with the group {$\rho_h$ : h ∈ $H_0$}), diam $\Gamma_{p,q}$ = max(ecc(I, A), ecc(c, B)). Since ψ($S_p$) is closed under inversion (Lemma 2.4), the side swap σ(g, A) = (g, B), σ(g, B) = (g, A) is an automorphism of H(PGL(2, q), ψ($S_p$)) ([H] Lemma 3.1(iv)); and $\rho_c$ is one too. $\rho_c$∘σ maps (I, A) to (c, B) and maps $\Gamma_{p,q}$ onto the component of (c, B), which is $\Gamma_{p,q}$ since (c, B) ∼ (I, A). Hence ecc(c, B) = ecc(I, A). (d) The map (g, A) ↦ (g, A), (h, B) ↦ (c⁻¹h, B) is the isomorphism H(G, S) → H(G, c⁻¹S) of [H] Lemma 3.1(i); restricted to $\Gamma_{p,q}$ it lands in PSL(2, q) × {A, B}, and both graphs are Δ-regular with q(q² − 1) vertices. ■

### 2.2 The order-3 symmetry

**Theorem 2.6.** Let p, q be odd primes, q > p, let ζ = 1 + i + j + k and h = ψ(ζ).

(a) h ∈ GL(2, q) has determinant 4, so [h] ∈ PSL(2, q), and [h] has order 3 in PGL(2, q).
(b) For every α ∈ H(Z), hψ(α)h⁻¹ = ψ(σ(α)), where σ($a_0$ + $a_1$i + $a_2$j + $a_3$k) = $a_0$ + $a_3$i + $a_1$j + $a_2$k.
(c) Conjugation by h permutes the p + 1 points of ψ($S_p$). Its fixed points are the [ψ(α)] with $a_1$ = $a_2$ = $a_3$, and its cycle type is

| p | cycle type on ψ($S_p$) | Δ = p + 1 | records with this cycle type ([H] §5.1) |
|---|---|---|---|
| p = 3 | 1 + 3 | 4 | none (Δ = 4 records are metacyclic) |
| p ≡ 1 (mod 3) | $1 + 1 + 3^{(p-1)/3}$ | 8 (p = 7), 14 (p = 13), 20, 32, 38, … | (8,8), (8,9), (8,10); (14,9), (14,10) |
| p ≡ 2 (mod 3) | $3^{(p+1)/3}$ | 6 (p = 5), 12 (p = 11), 18, 24, 30, … | (6,9), (6,10); (12,9), (12,10) |

So the LPS Haar graph is an orbit-closed Haar graph in the sense of [H] Construction 3.2, with α = conjugation by an element of order 3 of PSL(2, q) and m = 1, and with the same cycle type as the records at Δ = 6, 8, 12 and 14.

Proof. (a) $N(\zeta) = 4$, so $\det h = 4 = 2^2$, a nonzero square ($q$ odd). Write $v = i + j + k$, so $v^2 = -3$ and $\zeta = 1 + v$; then $\zeta^2 = 1 + 2v + v^2 = -2 + 2v$ and $\zeta^3 = (-2 + 2v)(1 + v) = -2 - 2v + 2v + 2v^2 = -8$. So $h^3 = -8I$ is a scalar. And $h$ is not a scalar: $\psi(\zeta) - \tilde{c}I = \psi(\zeta - \tilde{c})$ would vanish only if $\zeta - \tilde{c} \in qH(Z)$ (Lemma 2.3), but the i-coordinate of $\zeta - \tilde{c}$ is 1. So [h] has order 3.

(b) We compute $\zeta i \zeta$. First $\zeta i = i + i^2 + ji + ki = -1 + i + j - k$. Then $(-1 + i + j - k)(1 - i - j - k) = (-1 + i + j + k) + (i + 1 + j - k) + (j + k + 1 - i) + (-k + j - i - 1) = 4j$, using $ij = k$, $ik = -j$, $ji = -k$, $jk = i$, $ki = j$, $kj = -i$. So $\zeta i \zeta = 4j$, that is, $\zeta i \zeta^{-1} = j$. Applying the ring automorphism of $H(Z)$ that permutes $i \to j \to k \to i$ (it preserves the relations and fixes $\zeta$), we get $\zeta j \zeta^{-1} = k$ and $\zeta k \zeta^{-1} = i$. Hence $\zeta \alpha \zeta^{-1} = \sigma(\alpha)$, and $\zeta \alpha \zeta = 4\sigma(\alpha)$. Applying $\psi$ (Lemma 2.3): $h\psi(\alpha) \cdot \mathrm{adj}(h) = 4\psi(\sigma(\alpha))$, and $\mathrm{adj}(h) = \det(h) \cdot h^{-1} = 4h^{-1}$.

(c) $\sigma$ preserves the norm, the coordinate $a_0$ and the parity pattern of $(a_1, a_2, a_3)$. If $p \equiv 1 \pmod 4$, $\sigma(S_p) = S_p$. If $p \equiv 3 \pmod 4$, $\sigma(\alpha)$ satisfies Definition 2.1 except possibly the sign rule when $a_0 = 0$, so $\sigma(\alpha)$ or $-\sigma(\alpha)$ lies in $S_p$, and $\psi(-\beta) = -\psi(\beta)$ is the same point of $PGL(2, q)$. So conjugation by $h$ permutes $\psi(S_p)$. By Lemma 2.4(a), $[\psi(\sigma\alpha)] = [\psi(\alpha)]$ iff $\sigma(\alpha) = \pm\alpha$. The case $\sigma(\alpha) = -\alpha$ forces $a_0 = 0$ and $(a_3, a_1, a_2) = (-a_1, -a_2, -a_3)$, so $a_1 = -a_3 = a_2 = -a_1$, so $\alpha = 0$, impossible. So the fixed points are the $\alpha \in S_p$ with $a_1 = a_2 = a_3 =: a$, i.e. with $a_0^2 + 3a^2 = p$.

- If $p \equiv 2 \pmod 3$, $a_0^2 + 3a^2 \equiv a_0^2 \in \{0, 1\} \pmod 3$, so there is no fixed point.
- If $p \equiv 1 \pmod 3$, the equation $a_0^2 + 3a^2 = p$ has exactly four integer solutions $(\pm a_0, \pm a)$, with $a_0, a \neq 0$ (Fermat; for a prime $p \equiv 1 \bmod 3$ the representation $p = x^2 + 3y^2$ with $x, y > 0$ is unique; cited: D. A. Cox, Primes of the form $x^2 + ny^2$, Wiley 1989, §1 [secondary]). If $p \equiv 1 \pmod 4$ then $a_0$ is odd and $a$ even (odd + odd $\equiv 0$ and even + $3 \cdot$odd $\equiv 3 \bmod 4$), so exactly the two solutions with $a_0 > 0$, namely $a_0 + a(i + j + k)$ with $a = \pm|a|$, are in $S_p$. If $p \equiv 3 \pmod 4$ then $a_0$ is even and $a$ odd, $a_0 \neq 0$ ($p \neq 3a^2$), and again exactly the two with $a_0 > 0$ are in $S_p$. So there are two fixed points.
- If $p = 3$: $a_0 = 0$, $a = \pm 1$, and of $\pm(i + j + k)$ only $i + j + k$ is in $S_3$. One fixed point.

Since [h] has order 3, every other point lies in a 3-cycle, which gives the cycle types. ■

**Theorem 2.7 (cited: Lubotzky–Phillips–Sarnak, Combinatorica 8 (1988) 261–277 [secondary for the constants]; Davidoff–Sarnak–Valette (2003), Ch. 4 [secondary]).** Let $p \equiv 1 \pmod 4$ and let $q > p$ be an odd prime. Then $\Gamma_{p,q}$ has $q(q^2 - 1)$ vertices (that is, the LPS Cayley graph $X^{p,q}$ is connected), every eigenvalue of its adjacency matrix other than $\pm(p + 1)$ has absolute value at most $2\sqrt{p}$ (Ramanujan), and $\mathrm{diam}\, \Gamma_{p,q} \leq 2 \log_p n + O(1)$, with an explicit constant given by LPS.

Remark. $\Gamma_{p,q}$ is $X^{p,q}$ itself when $p$ is not a square mod $q$ ($X^{p,q}$ is then a bipartite Cayley graph of $PGL(2, q)$, and a bipartite Cayley graph whose connection set lies in the non-trivial coset of an index-2 subgroup is a Haar graph of that subgroup, Lemma 2.5(d)); it is the bipartite double cover of $X^{p,q}$ when $p$ is a square mod $q$. Computer-checked: $\Gamma_{p,q}$ has $q(q^2 - 1)$ vertices for $p \in \{3, 5, 7, 11, 13\}$ and every prime $p < q \leq 160$ (logs `haar_families/scan/lps_p*.txt`, column "vertices"), including $p \equiv 3 \pmod 4$, which Theorem 2.7 as cited does not cover.

### 2.3 The family written out at Δ = 6

**Example 2.8 (p = 5).** $S_5 = \{1 \pm 2i, 1 \pm 2j, 1 \pm 2k\}$. With $x^2 + y^2 = -1$ in $F_q$, the six matrices are

$$\psi(1 \pm 2i) = \begin{pmatrix} 1 \pm 2x & \mp 2y \\ \mp 2y & 1 \mp 2x \end{pmatrix}, \quad \psi(1 \pm 2j) = \begin{pmatrix} 1 & \pm 2 \\ \mp 2 & 1 \end{pmatrix}, \quad \psi(1 \pm 2k) = \begin{pmatrix} 1 \pm 2y & \pm 2x \\ \pm 2x & 1 \mp 2y \end{pmatrix},$$

and conjugation by $h = \psi(1 + i + j + k) = [[1 + x + y, 1 + x - y], [-1 + x - y, 1 - x - y]]$ cycles $1 + 2i \to 1 + 2j \to 1 + 2k \to 1 + 2i$ and the same with $-2$: cycle type $3 + 3$. In the coordinates of §1.2, **the point [a : b : c : d] of the first copy of PSL(2, q) is joined to the six points**

$$[\,(1 \pm 2x)a \mp 2yc : (1 \pm 2x)b \mp 2yd : \mp 2ya + (1 \mp 2x)c : \mp 2yb + (1 \mp 2x)d\,],$$

$$[\,a \pm 2c : b \pm 2d : c \mp 2a : d \mp 2b\,], \qquad [\,(1 \pm 2y)a \pm 2xc : (1 \pm 2y)b \pm 2xd : \pm 2xa + (1 \mp 2y)c : \pm 2xb + (1 \mp 2y)d\,]$$

of the second copy; and the point [a : b : c : d] of the second copy is joined to the same six expressions, read as points of the first copy (by Lemma 2.3, adj $\psi(\alpha) = \psi(\bar{\alpha})$, and $\bar{S}_5 = S_5$, so the six adjugates are the same

six matrices up to scalars). For q = 13, with (x, y) = (0, 5) (the first solution of $x^2 + y^2 + 1 \equiv 0$ found by the programs), the component of the first copy's identity is a 6-regular bipartite graph on 2184 vertices of diameter 7 (computer-checked by `lps_haar`, `haar_ecc` and the all-pairs BFS `apsp.c`, §8).

---

## 3. The 4/3 obstruction

**Lemma 3.1 (walk values).** Let p, q be odd primes, q > p. A walk of even length L in H(PGL(2, q), ψ($S_p$)) from (I, A) ends at a vertex ([ψ(β)], A) with β ∈ H(Z), N(β) = $p^L$ and β ≡ 1 (mod 2H(Z)); in particular the real coordinate $b_0$ of β is odd.

Proof. The walk with letters ψ($\alpha_1$), ..., ψ($\alpha_L$) ends at $s_L{}^{-1}$s_{L−1}⋯$s_2{}^{-1}$s_1 with $s_t$ = ψ($\alpha_t$), and [ψ(α)$^{-1}$] = [ψ($\tilde{\alpha}$)] (Lemma 2.3). So it ends at [ψ(β)] with β = $\tilde{\alpha}_L\ \alpha_{L-1} \cdots \tilde{\alpha}_2\ \alpha_1$, and N(β) = $p^L$. Reduce mod 2: in H(Z)/2H(Z) we have −1 = 1, so i, j, k commute there. If p ≡ 1 (mod 4), every $\alpha_t$ and $\tilde{\alpha}_t$ is ≡ 1 (mod 2H(Z)), so β ≡ 1. If p ≡ 3 (mod 4), every $\alpha_t$ and $\tilde{\alpha}_t$ is ≡ v := i + j + k, and $v^2 = i^2 + j^2 + k^2 + 2(ij + jk + ki) \equiv 1$, so β ≡ $v^L$ = 1 since L is even. ■

**Theorem 3.2 (distance to ψ(i)).** Let p, q be odd primes with q > p, and let $L_0$ be the least even integer L ≥ 0 with $2p^{L/2} > q^2$. In $\Gamma_{p,q}$:

(a) d((I, A), ([ψ(i)], A)) ≥ $L_0$, and the same holds for ψ(j) and ψ(k);
(b) if p ≡ 1 (mod 4), equality holds.

Note that det ψ(i) = N(i) = 1, so [ψ(i)] ∈ PSL(2, q) for every q.

Proof. (a) A path from (I, A) to ([ψ(i)], A) has even length L (the graph is bipartite). By Lemma 3.1 there is β with N(β) = $p^L$, $b_0$ odd and [ψ(β)] = [ψ(i)], i.e. ψ(β) = cψ(i) with c ∈ $F_q$^×. By Lemma 2.3, β − $\tilde{c}$i ∈ qH(Z), so q | $b_0$, $b_2$, $b_3$. Put P = $p^{L/2}$, an integer. Since $b_0$ is odd, $b_0 \neq 0$, so $b_0^2 \geq q^2$, and then $b_1^2 = P^2 - b_0^2 - b_2^2 - b_3^2 \leq P^2 - q^2$, so $|b_1| < P$. Also $q^2$ divides $b_0^2 + b_2^2 + b_3^2 = P^2 - b_1^2 = (P - b_1)(P + b_1)$. The prime q does not divide P (q ≠ p), so q cannot divide both factors (it would divide their sum 2P). Hence $q^2$ divides P − ε$b_1$ for some ε ∈ {1, −1}. Since $|b_1| < P$, $0 < P - \varepsilon b_1 < 2P$, so $q^2 \leq P - \varepsilon b_1 < 2P$. That is $2p^{L/2} > q^2$, and L ≥ $L_0$. For ψ(j) and ψ(k): conjugation by h fixes (I, A), is an automorphism of H(PGL(2, q), ψ($S_p$)) (Theorem 2.6 and [H] Prop. 3.3(a)), and maps [ψ(i)] to [ψ(j)] to [ψ(k)] (Theorem 2.6(b)), so the three distances are equal.

(b) Let L = $L_0$, P = $p^{L/2}$, and put $n_1 = 2P - q^2 > 0$. Here P ≡ 1 (mod 4) (p ≡ 1 mod 4), so 2P ≡ 2 (mod 8), and $q^2 \equiv 1$ (mod 8), so $n_1 \equiv 1$ (mod 8). By the Gauss–Legendre three-square theorem (cited; n is a sum of three squares unless $n = 4^a(8b + 7)$), $n_1 = u^2 + v^2 + w^2$ with integers u, v, w; since $n_1 \equiv 1$ (mod 4), exactly one of them is odd, and we name it u. Put

$$\beta = qu + (P - q^2)\, i + qv\, j + qw\, k.$$

Then $N(\beta) = q^2(u^2 + v^2 + w^2) + (P - q^2)^2 = q^2(2P - q^2) + P^2 - 2Pq^2 + q^4 = P^2 = p^L$; $b_0 = qu$ is odd, $b_1 = P - q^2$ is even, and $b_2 = qv$, $b_3 = qw$ are even; so β ≡ 1 (mod 2H(Z)). And β ≡ P·i (mod qH(Z)), so [ψ(β)] = [Pψ(i)] = [ψ(i)]. By the LPS factorization theorem (cited: LPS 1988 §3; Davidoff–Sarnak–Valette (2003), Ch. 2, the unique factorization of quaternions β ≡ 1 mod 2 of norm $p^L$ [secondary]), $\beta = \pm p^r\ \alpha_1\alpha_2\cdots\alpha_{L-2r}$ with $\alpha_t \in S_p$ and r ≥ 0. Since [ψ(±$p^r$ γ)] = [ψ(γ)], we get [ψ(i)] = [ψ($\alpha_1\cdots\alpha_{L'}$)] with L′ = L − 2r even. Choose the letters $s_1$ = ψ($\alpha_{L'}$), $s_2$ = ψ($\tilde{\alpha}_{L'-1}$) (so [$s_2{}^{-1}$] = [ψ($\alpha_{L'-1}$)]), $s_3$ = ψ($\alpha_{L'-2}$), $s_4$ = ψ($\tilde{\alpha}_{L'-3}$), ...; they lie in ψ($S_p$) up to sign (Lemma 2.4(b)), and the A-word $s_{L'}{}^{-1} \cdots s_2{}^{-1}s_1$ has value [ψ($\alpha_1\cdots\alpha_{L'}$)]. So there is a walk of length L′ ≤ $L_0$ from (I, A) to ([ψ(i)], A). With (a), the distance is $L_0$. ■

**Corollary 3.3.** For odd primes q > p, with n = q($q^2$ − 1) ≥ |V($\Gamma_{p,q}$)|:

$$\operatorname{diam}\Gamma_{p,q} \;\geq\; L_0 \;\geq\; 4\log_p q - 2\log_p 2 \;>\; \tfrac{4}{3}\log_p n - 2\log_p 2.$$

Proof. $2p^{L_0/2} > q^2$ gives $L_0 > 2\log_p(q^2/2) = 4\log_p q - 2\log_p 2$, and $n < q^3$. ■

**Corollary 3.4 (a vanishing fraction of the Moore bound).** Let p be an odd prime, D ≥ 2, and q > p a prime with diam $\Gamma_{p,q} \leq D$. Then $q^2 < 2p^{\lfloor D/2 \rfloor}$, the number of vertices satisfies

$$n \le q(q^2-1) < 2^{3/2}\, p^{\frac{3}{2}\lfloor D/2 \rfloor} \le 2^{3/2} p^{3D/4},$$

and n < √2·$p^{1-D/4}$·$M^b$(p + 1, D). In particular, along the LPS family the fraction of the Moore bound tends to 0 both for fixed p and D → ∞ and for fixed D ≥ 5 and p → ∞.

Proof. $L_0$ ≤ diam ≤ D and $L_0$ is even, so $L_0$ ≤ 2⌊D/2⌋ and $2p^{\lfloor D/2 \rfloor} \ge 2p^{L_0/2} > q^2$. Then n < $q^3$ < $(2p^{\lfloor D/2 \rfloor})^{3/2}$. Finally $M^b$(p + 1, D) = $2\Sigma_{j<D}\ p^j > 2p^{D-1}$, so n/$M^b$ < $2^{3/2}p^{3D/4}/(2p^{D-1})$ = √2·$p^{1-D/4}$. ■

For comparison: at D = 8 the bound is √2/p (28 % for p = 5), and the LPS graphs actually reach 1–6 % (Appendix B); the lab's records at D = 8 are at 8–22 %, and the D = 8 table values with Δ ≥ 10 (Gómez's compounds of generalized hexagons) at about 10 % ([H] §5, §6). In the regime D fixed, Δ → ∞, Delorme's constructions from generalized polygons have lim inf n·$\Delta^{1-D}$ > 0 for every D (C. Delorme, J. Graph Theory 9 (1985) 325–334 [abstract]), so arithmetic Haar families lose to them by a factor $p^{D/4-1}$.

**Computer-checked 3.5.** For p ∈ {3, 5, 7, 11, 13, 17, 19, 29} and every prime q with p < q ≤ 160, the distance from (I, A) to ([ψ(i)], A), ([ψ(j)], A) and ([ψ(k)], A), computed by breadth-first search (`lps_haar.c`), equals $L_0$: no exception in [[$N_{THMB}$]] cases (log `haar_families/recheck_thmB.log`). This includes p ≡ 3 (mod 4), where equality is not proved here. The eccentricity of (I, A) exceeds $L_0$ by 0, 1 or 2 ([[ECC_DIST]]). So in this range the special vertex ψ(i) alone accounts for the diameter up to at most 2. An independent re-implementation by squad card c8 (its own quaternion arithmetic and BFS; `research/squads/thmfam-r1/c8/`) confirmed Theorem 2.6(c) for all 137 pairs with 7 ≤ q ≤ 61 and Theorem 3.2 in 41 cases with q ≡ 1 (mod 4), and its all-pairs BFS (`apsp.c`) agreed with `haar_ecc` on six LPS graphs. (The squad card also found a bug in the first version of `lps_haar.c`, which mixed the distances of the two sides; the logs above are from the corrected version.)

**Remark 3.6 (prior art).** The lower bound (4/3) log_p n + O(1) for the diameter of the bipartite LPS graphs is due to N. T. Sardari, Diameter of Ramanujan graphs and random Cayley graphs, Combinatorica 39 (2019) 427–446, arXiv:1511.09340 [abstract]; his abstract also states that almost all pairs of vertices of a Ramanujan graph are at distance at most (1 + ε) $log_p$ n, and reports numerics in which LPS diameters grow like (4/3) log and those of random Cayley graphs like log. We did not read his proof; the argument above is short and may be his. The exact value in Theorem 3.2(b) follows from the same arithmetic and the LPS factorization; we did not check whether it is stated there. That the LPS sets are permuted by the Hurwitz unit (1 + i + j + k)/2 is classical; we found no source that compares their cycle types with orbit-searched connection sets, and claim nothing more than the observation.

---

## 4. Cyclic factors

**Proposition 4.1.** Let K be a finite group, κ: $S_p$ → K any map, and S̃ = {(ψ(α), κ(α)) : α ∈ $S_p$} ⊆ PGL(2, q) × K. Then for every k ∈ K, the distance in H(PGL(2, q) × K, S̃) from (I, $e_K$, A) to ([ψ(i)], k, A) is at least $L_0$. Hence if the component of (I, $e_K$, A) has diameter ≤ D and contains ([ψ(i)], k, A) for some k, then $q^2 < 2p^{\lfloor D/2 \rfloor}$.

Proof. The projection (g, k, X) ↦ (g, X) maps every edge {(g, k, A), (ψ(α)g, κ(α)k, B)} to the edge {(g, A), (ψ(α)g, B)}, so it maps walks to walks of the same length and does not decrease distances. Apply Theorem 3.2(a). ■

So a cyclic factor cannot help at the vertex ψ(i) with a fixed q; what it changes is that m is free. The order is q(q² − 1)·m, the obstruction bounds only q, and the counting bound ([H] Thm. 4.1) allows m up to about $p^{D-1}/q^3$. Whether the cyclic part can be filled efficiently is a computational question:

**Computer-checked 4.2 (squad cards c6, c7).** Let r have order 3 mod m. Labels α ↦ z(α) ∈ $Z_m$ that are equivariant for the order-3 symmetry (z(σα) = r·z(α); fixed letters get 0; one 3-cycle gets (1, r, $r^2$) without loss of generality, the others (t, rt, $r^2$t)) make H(PGL(2, q) × $Z_m$, S̃) orbit-closed with α(x, z) = (hxh⁻¹, rz), exactly the records' shape. Searching t for each (q, m), the squads found:

| cell | best LPS × $Z_m$ (q, m) | vertices | of table | lab record | LPS alone (Appendix B) |
|---|---|---|---|---|---|
| (6,8) | (13, 13) | 28,392 | 95.3 % | 31,680 | 12,144 |
| (6,9) | (13, 43) | 93,912 | 80.0 % | 127,296 | 24,360 |

| cell | best LPS × $Z_m$ (q, m) | vertices | of table | lab record | LPS alone (Appendix B) |
|---|---|---|---|---|---|
| (6,10) | (23, 37) | 449,328 | 99.4 % | 524,160 | 148,824 |
| (8,8) | (11, 151) | 199,320 | 98.9 % | 239,904 | 29,760 |
| (8,9) | [[$C7_{89}$]] | | | 1,341,600 | 300,696 |

Every entry was re-checked with `haar_ecc` (two-source BFS on the component) and, for (6,10), with `verify_cover.py` (independent Python arithmetic; eccentricities 10 and 10). These graphs are below the lab's records and below the table, and their labels were searched for each (q, m): they are not a uniform family. They show that the cyclic factor removes most of the 4/3 loss in practice: at (6,10) the cover is 3.0 times the LPS graph alone.

---

## 5. Other uniform families (squad round 1)

**Score.** For a family S(q) and the primes 23 ≤ q ≤ 113, the excess is diam − $log_{\Delta-1}$|G| with |G| = one side of the graph, averaged over q (`haar_families/family_score.py`, which runs `haar_ecc`). Lower is better. A uniform family gets one S per q, so it should be compared with one random orbit-closed draw per q, not with the records, which are the best of millions of evaluations for one (q, m).

**Computer-checked 5.1 (mean excess, 22 primes 23 ≤ q ≤ 113, group PSL(2, q) unless stated; from `research/squads/thmfam-r1/scoreboard.jsonl`).**

| family | Δ | mean excess | random orbit-closed draw, same Δ | card |
|---|---|---|---|---|
| LPS, p = 5 (component of PGL(2, q)) | 6 | 4.00 | 3.18–3.50 (3 seeds) | director |
| integer lifts, O(X) ∪ O(Y), X, Y ∈ SL(2, Z), entries ≤ 3 (best of 206) | 6 | 3.36 | 3.18–3.50 | c1 |
| integer lifts, {I} ∪ O(X) ∪ O(Y) (best of 1212; mean of top 10: 3.35) | 7 | 3.10 | 2.96–3.24 (5 seeds, mean 3.07) | c2 |
| primitive-root formulas, e.g. X = [[g, 1], [0, $g^{-1}$]], Y = [[1/2, g], [g, 2(1 + $g^2$)]] (best of 16) | 7 | 2.96 | 2.96–3.24 | c10 |
| Dickson / trace / power formulas (best of 15) | 6 | 3.13 | 3.18–3.50 | c11 |
| conic model, Cayley transforms of integer matrices (best of 8) | 8 | 3.34 | 3.25–3.30 | c16 |

| family | Δ | mean excess | random orbit-closed draw, same Δ | card |
|---|---|---|---|---|
| order-5 or order-6 symmetry, single orbits (k = 6, best) | 12 | 2.73 | k = 3: 2.55–2.64; k = 6: 2.91–3.09 | c15 |
| LPS + one extra α-orbit (repair) | 9 | 3.24 (7 primes) | 2.72–2.86 | c18 |
| [[MORE_ROWS]] | | | | |
| the lab's records (one (q, m) each, annealed) | 6–16 | about 2.0–2.3 | | [H] |

Torus-orbit families with Δ growing like q (card c12: S = {I} ∪ T-conjugates of x for a torus T of PSL(2, q)) have diameter 5 to 7 for 31 ≤ q ≤ 61 and reach at most 1.8 % of the Moore bound at their diameter; no family of diameter ≤ 4 appeared. The excess of LPS grows with log_5|G| with slope 0.41 (card c14, 38 primes 23 ≤ q ≤ 199; Corollary 3.3 predicts 1/3), that of random orbit-closed draws with slope −0.04.

**Heuristic 5.2 (reading of 5.1).**

1. The best uniform formula of each kind ties a single random draw; it does not beat it by more than the spread of random seeds (about 0.1–0.2), and the "best" was selected among 8–1212 variants on the same primes.
2. Arithmetic structure costs: LPS is about 0.6–0.8 worse than one random draw at q ≤ 113, and the gap grows like (1/3) log_p n (Theorem 3.2, card c14).
3. Larger symmetry (k = 5, 6) is worse than k = 3 at the same Δ, for formulas and for random draws (card c15), in line with [H] §3.5: orbit closure helps the search, not the quality of a single set.
4. The records are about 1 below a single draw. In the model of [H] §3.5 a single random S misses about μ = N $e^{-W/N}$ elements and is valid with probability about $e^{-\mu}$; the search raises the usable μ to μ* with $e^{\mu*}$ ≈ the number of independent tries, which shrinks the needed W/N by ln μ*, about one unit of $log_{\Delta-1}$. A formula gives one try per q.

---

## 6. Why no record-quality uniform family is in sight

**6.1 Fixed Δ, D → ∞.** By Proposition 1.1 a family here has q → ∞. A uniform family of record quality would have diam ≤ $log_{\Delta-1}$ n + O(log log n) along an explicit sequence of groups. No explicit family of bounded-degree graphs with diam ≤ (1 + ε) $log_{\Delta-1}$ n is known [secondary]; random Δ-regular graphs have diameter (1 + o(1)) $log_{\Delta-1}$ n (B. Bollobás, W. Fernandez de la Vega, Combinatorica 2 (1982) 125–134 [secondary]), and so, conjecturally, do random Haar graphs of SL(2, q) (research/theory/THRESHOLD.md, Conjecture 6, on master). The best explicit candidates, the Ramanujan graphs, have almost all distances (1 + o(1)) $log_{\Delta-1}$ n (Sardari [abstract]; E. Lubetzky, Y. Peres, Cutoff on all Ramanujan graphs, GAFA 26 (2016) [secondary]), but for the LPS graphs a few vertices are at distance (4/3) log_p n (Theorem 3.2). That is the precise way in which the most natural uniform family of the records' shape fails, and Corollary 3.4 turns it into a loss of a factor $p^{D/4-1}$ against the Moore bound.

**6.2 Fixed D, Δ → ∞.** The LPS family with q ≈ $p^{D/4}$ has degree p + 1 and order about $p^{3D/4}$ at diameter ≈ D (Corollary 3.4 is an upper bound; the lower side is Theorem 2.7, order $p^{(D-O(1))/2}$). Delorme's polygon constructions reach a fixed fraction of $p^{D-1}$ (§3). We know of no Haar-graph family of the records' groups that reaches lim inf n·$\Delta^{1-D}$ > 0 at D = 8, 9 or 10; the random-S theorem of research/theory/THRESHOLD.md reaches n ≥ $\Delta^{D-1}$/((D − 1)!·(1 + η)·ln n) for every finite group, which loses a logarithm.

**6.3 What would change this.** A formula that beats one random draw by about 1 in excess, uniformly in q; or a proof that some explicit cover (for example LPS × $Z_m$ with labels given by a rule) has diameter log_p n + O(log log n). Neither is in sight. The squads' round 2 looked for label rules for LPS × $Z_m$ (Appendix A).

## 7. Prior art

- **LPS graphs.** A. Lubotzky, R. Phillips, P. Sarnak, Ramanujan graphs, Combinatorica 8 (1988) 261–277; G. A. Margulis, Problems Inform. Transmission 24 (1988) 39–46 [secondary]; the book of Davidoff–Sarnak–Valette (2003) [secondary]. Definitions 2.1–2.2, Lemma 2.3, Theorem 2.7 and the factorization used in Theorem 3.2(b) are theirs.
- **The 4/3 lower bound.** Sardari (2019) [abstract], Remark 3.6.
- **Typical distances in Ramanujan graphs.** Lubetzky–Peres (2016) [secondary]; Sardari (2019) [abstract].
- **Haar graphs of SL(2, q) and PSL(2, q) in extremal problems.** [H] §6.1: Exoo–Jajcay (girth, a Haar graph of SL(2, 83)); Bray–Parker–Rowley (the group PSL(2, q) × $Z_3$); the LPS graphs themselves are Haar graphs of PSL(2, q) in the bipartite case (Lemma 2.5(d)). In the bipartite degree–diameter table at D = 8–10 (Miller–Širáň, DS14v2, 2013, as reproduced by the Combinatorics Wiki) we found no LPS graph; the novelty check `research/lit/novelty-2026-10-25.md` (branch `claude/novelty-literature-check-4i5frp`, claim 5) found no Haar graphs of SL(2, q) or PSL(2, q) there.
- **Asymptotic lower bounds for $N^b(\Delta, D)$.** Delorme (1985) [abstract via the novelty check]; random bipartite graphs (V. Klee, D. Larman, E. M. Wright, The diameter of almost all bipartite graphs, Studia Sci. Math. Hungar. 15 (1980) 39–43 [secondary]); research/theory/THRESHOLD.md for random Haar graphs.
- **Orbit-closed connection sets.** [H] §6.1 (an old device; Singer sets and their multipliers).

What this note adds: Theorem 2.6 (the LPS sets have the records' orbit shape, with the cycle type computed), the elementary proof and the exact value in Theorem 3.2, Corollary 3.4 in the form of a fraction of the Moore bound, the covers of §4 and the measurements of §5. We claim no novelty for Lemmas 2.3–2.5 or for Theorem 2.7.

---

## 8. How to re-check

```
cd research/theorems/haar_families
cc -O3 -o lps_haar lps_haar.c -lm      # LPS graph: generators (-v), eccentricities, d(I, psi(i)) etc.
cc -O3 -o haar_ecc haar_ecc.c          # any Haar graph of SL/PSL/PGL(2,q) x Z_m from generator lines "a b c d z"
./lps_haar 5 13 -v                     # Example 2.8: 2184 vertices, diameter 7, d(I, psi(i)) = 6 = L0
python3 check_thmB.py 3,5,7,11,13,17,19,29 160 > recheck_thmB.log     # Computer-checked 3.5
./lps_haar 5 13 -v | grep '^gen' | sed 's/.*mat(//; s/) z=/ /; s/,/ /g' > g.txt
python3 component_edges.py pgl 13 1 e.txt < g.txt                    # independent edge list of the component
cc -O2 -o apsp ../../lifts/coordinator_check/apsp.c && ./apsp e.txt 6 40   # all-pairs BFS: diameter 7
python3 table_lps.py                   # Appendix B from scan/lps_p*.txt
python3 family_score.py --gen "./gen_lps.sh 5 {q}" --group pgl --qrange 23 113   # §5, LPS row
python3 ../../squads/thmfam-r1/c6/gen_c6.py 23 37 19 0 | ./haar_ecc pgl 23 37  # §4.2, (6,10)
python3 ../../squads/thmfam-r1/c6/gen_c6.py 23 37 19 0 | python3 verify_cover.py 23 37
```

---

## Appendix A. The squad rounds

[[SQUADS]]

## Appendix B. The LPS family cell by cell

[[APPB]]